value with probability 0.55. Scenario 2 has probability 0.25 and will result in recovery of USD0.50 per USD1 principal value with probability 0.85, or in recovery of USD0.40 per USD1 principal value with probability 0.15.

Using the data for Scenario 1 and Scenario 2, calculate the following:

A. Compute the expected recovery, given the first scenario.

B. Compute the expected recovery, given the second scenario.

C. Compute the expected recovery.

D. Graph the information in a probability tree diagram.

E. Compute the probability of each of the four possible recovery amounts: USD0.90, USD0.80, USD0.50, and USD0.40.

Solution:

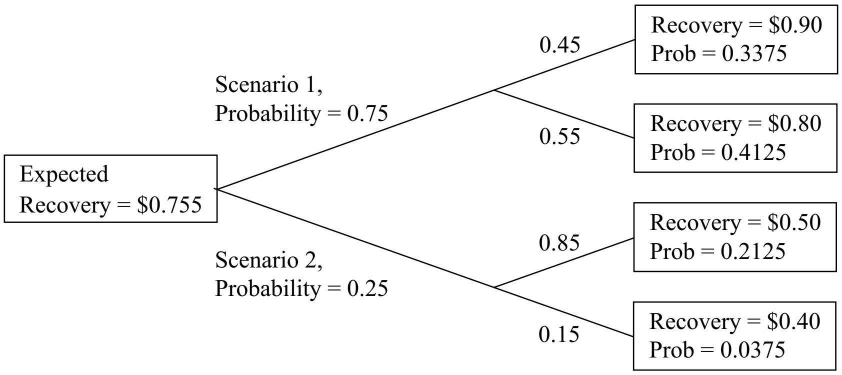
A. *Outcomes associated with Scenario 1:* With a 0.45 probability of a USD0.90 recovery per USD1 principal value, given Scenario 1, and with the probability of Scenario 1 equal to 0.75, the probability of recovering USD0.90 is 0.45 (0.75) = 0.3375. By a similar calculation, the probability of recovering USD0.80 is 0.55(0.75) = 0.4125.

Outcomes associated with Scenario 2: With a 0.85 probability of a USD0.50 recovery per USD1 principal value, given Scenario 2, and with the probability of Scenario 2 equal to 0.25, the probability of recovering USD0.50 is 0.85(0.25) = 0.2125. By a similar calculation, the probability of recovering USD0.40 is 0.15(0.25) = 0.0375.

B. $E(recovery \mid Scenario\ 1)$ = 0.45(USD0.90) + 0.55(USD0.80) = USD0.845

C. $E(recovery \mid Scenario\ 2)$ = 0.85(USD0.50) + 0.15(USD0.40) = USD0.485

D. $E(recovery)$ = 0.75(USD0.845) + 0.25(USD0.485) = USD0.755



BAYES' FORMULA AND UPDATING PROBABILITY ESTIMATES 4

☐ calculate and interpret an updated probability in an investment setting using Bayes' formula

A topic that is often useful in solving investment problems is Bayes' formula: what probability theory has to say about learning from experience.